ESP32 Power Check Executive CONOPS Brief

Mission

ESP32 Power Check supplies an inexpensive, unattended indication that power and the required communications path remain available at a monitored location. A USB-powered Seeed Studio XIAO ESP32-C3 sends an authenticated heartbeat to Gatus immediately after boot and every 30 seconds. Gatus changes endpoint health and delivers configured alerts when those heartbeats stop.

This is a continuity monitor, not an electrical meter. Loss of heartbeat may mean loss of monitored power, Wi-Fi, DNS, routing, time service, Gatus, or the device itself. Operations must correlate an alert with facility and network evidence before assigning root cause.

Operating concept



The installer provisions one to three 2.4 GHz Wi-Fi networks and a matching Gatus external endpoint. The device connects to the strongest available configured network, obtains time when needed for HTTPS, and sends outbound-only heartbeat requests. There is no inbound web server.

Gatus, rather than the device, owns alert thresholds and delivery. The supplied example waits 90 seconds without a push before the endpoint becomes unhealthy and applies additional failure/success thresholds to reduce alert flapping.

Resilience and visibility

The design is intended to run without routine intervention:

- disconnected Wi-Fi is retried every 10 seconds;
- after 3 failed reconnects, the device also tries configured networks in order;
- after 30 failed Wi-Fi attempts, it restarts itself;
- an invalid clock is retried through NTP every 60 seconds, and verified HTTPS waits for valid time rather than weakening certificate checks;
- non-`200` Gatus responses and transport errors are retried at the heartbeat cadence, with restart after 20 consecutive failures; and
- USB serial records boot count, reset reason, network association, time/TLS problems, response status, uptime, signal strength, IP address, and free heap.

At nominal intervals, the restart thresholds correspond to roughly five minutes of Wi-Fi join failures and ten minutes of heartbeat failures. Connection and request timeout duration can extend those elapsed times.

Security posture

Production deployments should use HTTPS with a trusted root CA and a DNS name that matches the certificate. The firmware authenticates to Gatus with a bearer token. Wi-Fi credentials, the token, and trust material reside in an ignored local secrets header and must not enter source control or operational tickets.

An option to disable TLS validation exists strictly for testing and emits a warning. It is not an acceptable production configuration. Physical access to the device and its USB maintenance interface must also be controlled.

Ownership and response

Owner	Accountability
System owner	Defines the monitored circuit, alert objective, and accepted residual risk.
Installer/maintainer	Provisions, flashes, validates, rotates credentials, and diagnoses serial output.
Network/Gatus administrator	Maintains Wi-Fi, DHCP, DNS/NTP, routing, endpoint configuration, and alert delivery.
Operator/responder	Monitors Gatus, verifies power first, then checks communications and service health.

Maintenance is triggered by credential or CA changes, network changes, Gatus migration, repeated restarts, TLS errors, or firmware updates. Retirement requires disabling the Gatus endpoint and alerts, revoking the token, removing the device, and updating asset records.

Success measures

An accepted installation:

- starts cleanly and reports its boot/reset information over 115200-baud serial;
- joins the intended 2.4 GHz network and obtains a valid address;
- validates time and certificates when using HTTPS;
- receives HTTP `200` from the correct Gatus endpoint at the 30-second cadence;
- becomes unhealthy and alerts according to policy when heartbeats cease;
- recovers when Wi-Fi or Gatus availability returns; and
- stores no production secret in tracked project files.

Detailed interfaces, timings, operational scenarios, fault response, and acceptance procedures are defined in `CONOPS.md`.